SARANG AUTE

B.E.
Electronics & Telecommunication Engineering
JSPM Narhe Technical Campus, Pune

+91-7448089562
sarangsaute03@gmail.com
https://github.com/SarangAute
linkedin.com/in/sarang-aute-936ba9271

Summary

– Motivated and passionate Firmware Engineer with strong expertise in Embedded C/C++, ARM7 architecture, and low-level driver development. Hands-on experience in UART, SPI, I2C, CAN, interrupt handling, and RTOS fundamentals, along with practical implementation of GSM communication and IoT-based systems using ESP32 and Raspberry Pi. Seeking an opportunity to design and develop efficient, scalable firmware solutions for real-world embedded applications.

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.E.	JSPM Narhe Technical Campus, Pune	8.29	2025
Class XII (HSC)	ST. Lawrence Super 30, Chh. Sambhajinagar	92.00%	2021
Class X (SSC)	Chate School, Chh. Sambhajinagar	67.00%	2019

Projects

- **Smart Waste Management with Segregation and Smart Notification Sponsored by Indian Railway.** *Apr 2024 - May 2025*
  Pune, Maharashtra
  – IoT-based Smart Waste Management System | ESP32, Raspberry Pi, Python, Embedded C Developed an automated waste segregation with 70% accuracy and notification system using ESP32 (TX) and Raspberry Pi (RX)with 100% accuracy; implemented real-time monitoring for centralized control; sponsored by Indian Railways.
- **Digital Dice with Dice History.** *Dec 2025*
  Chennai, Tamil Nadu
  – Developed an embedded-based digital dice system that generates pseudo-random numbers (1–6) on a button press and stores the last 10 roll values with 100% accuracy. Implemented roll history tracking and display using Embedded C for fair and reliable electronic gameplay with zero data loss during 200+ test iterations.
- **OTP Generation for High Security with SMS Alert.** *Jan 2026*
  Chennai, Tamil Nadu
  – Designed and developed a two-level authentication system using *Embedded C* on ARM7, integrating *GSM SIM800C (UART, AT commands)* for real-time OTP delivery and SMS alerts <3 second SMS delivery time and ensuring 100% OTP validation success rate. Implemented password verification, random OTP generation, and interrupt-driven IR sensor counting using *External Interrupt (EINT0)* for event-based notifications. Focused on secure access control, serial communication, and hardware–software integration.
- **CAN based Intelligence Accident Prevention with automatic Breaking & Airbag Activation System.** *Feb 2026*
  Chennai, Tamil Nadu
  – Designed and developed a multi-ECU CAN-based Intelligent Bus Accident Prevention System using Embedded C on ARM7, integrating 4 distributed ECUs for real-time vehicle safety control. Implemented ultrasonic sensor interfacing to measure obstacle distance and developed threshold-based automatic braking logic using PWM for dynamic brake actuation. Configured CAN protocol for reliable inter-ECU communication and implemented SPI for peripheral interfacing. Developed collision detection algorithm to trigger automatic airbag deployment during impact conditions. Focused on real-time processing, automotive communication protocols (CAN, SPI), low-level driver development, and safety-critical embedded firmware design.
- **Student Record Management System (C Programming).** *Mar 2026*
  Chennai, Tamil Nadu
  – Developed a menu-driven Student Record Management System using C, implementing singly linked lists for dynamic data storage and efficient record management. Designed modular functions to add, delete, search, modify, and display student records using pointer manipulation and dynamic memory allocation. Implemented sorting algorithms based on student name and marks to improve data organization. Integrated file handling mechanisms to store and retrieve records from persistent storage. Focused on data structures, algorithm implementation, memory management, and structured C programming.

Skills

- **Programming Languages**: C Programming | C++ | JAVA | Embedded C .
- **Operating System**: Windows | linux
- **Embedded & Firmware**: Bare-metal programming | Interrupt Handling.
- **Microcontrollers**: ARM 7 | LPC2129 | ESP32 | NodeMCU | Raspberry Pi.
- **Protocols**: UART | I2C | SPI | CAN.
- **Software tools**: Proficient in using Keil | Proteus | Arduino IDE | VSCode | PCB Design | GitHub | CANoe.
- **Non Technical Skills**: Effective Time Management | Leadership | Adaptive Nature | Teamwork |

Emotional Intelligence.

- **Soft Skills**: Communication Skills |Team Collabration | Adaptability | Problem Solving.

RELEVANT COURSEWORK

Electronic Circuit | Electrical Circuit | Digital Circuit | Digital Communication | Signal & System | Principle of Communication System | OOPs | DSA | JAVA | Microcontroller | DBMS | Power Device & Circuit | Project Management | Cellular Network | Mathematics | Python | Logical Reasoning | Critical Reasoning | Data Interpretation

LEADERSHIP RESPONSIBILITIES

- **Treasurer**, ETSA Students Forum, JSPM Narhe Technical Campus *Oct, 2022*
  - Managed departmental budget and allocated resources for events
  - Ensured accurate and transparent financial documentation for audits and financial reviews and also worked closely with department members and faculty to ensure financial decisions aligned with organizational goals.

Course

- **Vector India Private Limited – Advanced Embedded Systems Training** *2025-26*
  - Trained in *ARM and 8051 microcontrollers* with Architecture and hands-on development in *Embedded C/C++*, peripheral interfacing, and interrupt-driven programming. Experienced with serial communication protocols including *UART, I2C, SPI, and CAN*, along with *RTOS* concepts (task scheduling, semaphores, mutex, inter-task communication). Familiar with *Embedded Linux* fundamentals, *TCP/IP* networking, and socket programming.
- **JAVA & DSA – Core2web** *2024*
  - Object-Oriented Programming (OOP), Arrays, Strings, Recursion, Linked List, Stack, Queue, Searching & Sorting algorithms, Time & Space Complexity analysis.

ACHIEVEMENTS

- **Intermediate & Elementary Exam Passed**, *2016*
  - ∗ Successfully passed the exam with **B Grade**, showcasing proficiency in fundamental art and creative skills.
- **Dr. Homi Bhabha Balvaidnyanik Competition**, *2018*
  - ∗ Successfully passed the prestigious Competition, demonstrating strong scientific aptitude and problem-solving skills.
- **Most Helpful Student Award**, *2025*
  - ∗ Awarded for exceptional kindness, generosity, and continuous support to peers; recognized for maintaining a positive and encouraging attitude.

INTERESTS

–**Travelling**: Always eager to explore new destinations and immerse in diverse cultures.

–**Gym**: Dedicated to maintaining physical fitness and mental well-being through regular exercise.

–**Photography**: Passionate about capturing the beauty of nature through photography, with a focus.